Statistical Physics, Fault-Tolerance, Subsystem Codes, and Quantum Monte Carlo



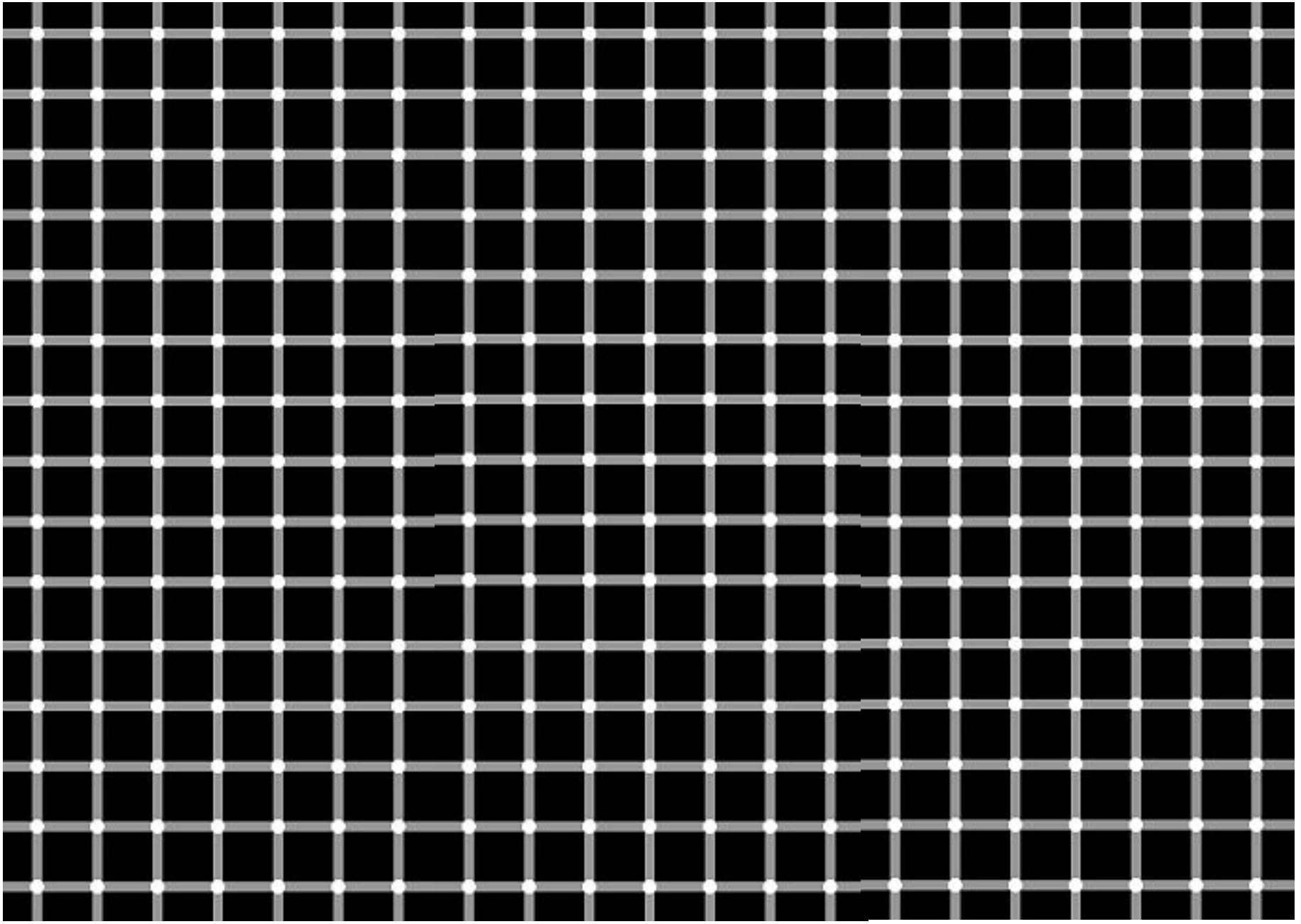
Rhapsody on a Theme by Kitaev



Dave Bacon

Preskill Group Meeting, April 24 2003

University of Queensland Quantum Annex
Right-side-Up Hemisphere



The Quantum Computing Roadmap



Th eria

QC Approach	QC Networkability	
	#6	#7
NMR		
Trapped Ion		
Neutral Atom		
Optical		
Solid State		
Superconductin		
Unique Qubits		
e-Helium		
Spectral Ho		

Legend: = a pote
= a pote
= no via

inciple

The column

1. A scala
2. The abi
3. Long (r
4. A unive
5. A qubit
6. The abi
7. The ability to faithfully transmit flying qubits between specified locations.

1941 – First programmable electronic calculator.
mechanical relays.
1943 – ENIAC.
18000 vacuum tubes: “nature abhors the vacuum tube.”
1947 – Bell labs develops the transistor.
1952 – G. W. Dummer proposes manufacturing electronic equipment in one block with no connecting wires.
1959 – Texas Instruments and Fairchild Semiconductor invent the integrated circuit.
1964 – first IBM 360 series (the concept of an “architecture”)
1964 – Integrated circuit which cost $1000 in 1959 now costs $10. Moore describes his law.
1972 – Intel 8008 and 8 bit microprocessor.
1975 – First personal computer the Altair.
1982 – IBM PC introduced. Personal computer revolution begins.

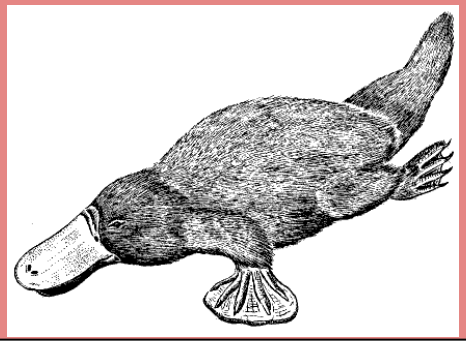
Have We Learned Anything?

History Only Teaches Me Physics

Interacting **Isolated**

What happens when many units (defined isolated) interact?

ordered
regular
reducible



turbulent
complex
not reducible

Platyputor

Computers are strange beasts in this world:

- built from parts which order and are regular
- reductionist's dream come true

yet:

- complexity of computer running a program is high
- algorithmic complexity implies reduction cannot be compressed beyond running the algorithm

The Physics Guarantee

What is the phase of matter corresponding to the computer?

PRACTICAL QUESTION (as opposed to philosophical)

There are distinct PHYSICAL reasons why robust classical computation is possible.

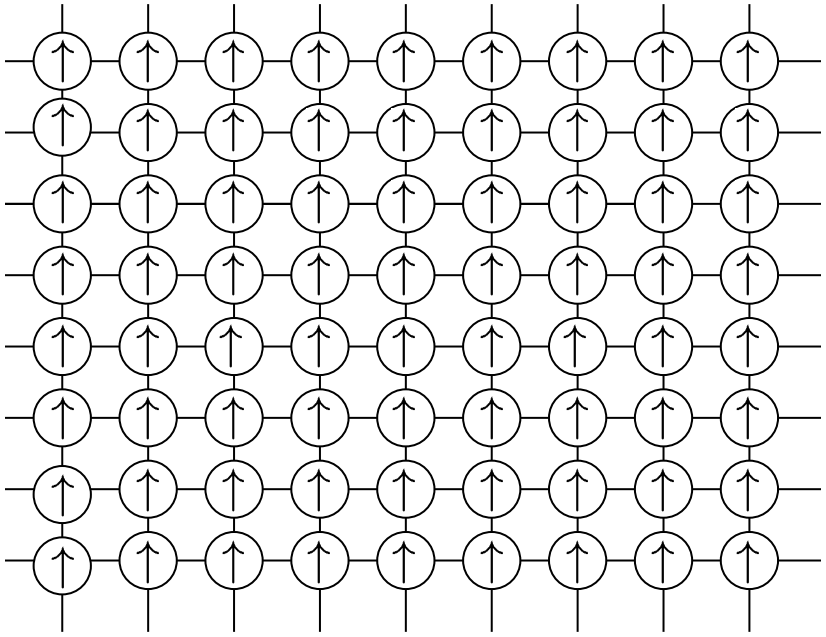
not all physical systems are equally good for computation:
there exists systems whose PHYSICS guarantees their ability to enact robust classical computation.

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

This talk: (1) this point of view for classical computers
(2) an attempt to port these ideas to a quantum computer

A Dynamic Ising Model





Ising model

spins on a lattice $s_i \in \{+1, -1\}$

$$E = -J \sum_{\langle ij \rangle} s_i s_j$$

Energetically favorable to align

Order parameter:

$$M = \sum_i s_i$$

(the information)

Dynamic Ising model

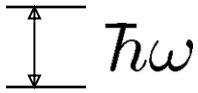
interaction with environment causes spins to flip



Reminder



two-level system radiatively coupled to a thermal reservoir:



$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

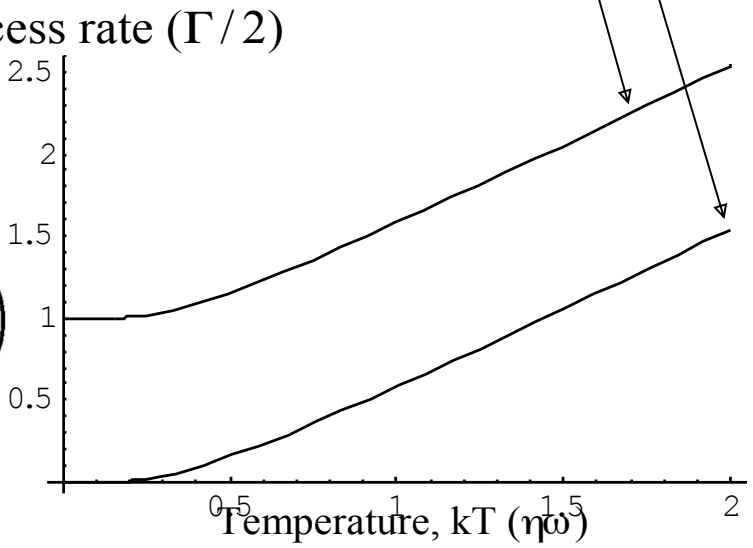
$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

"non-dissipative"

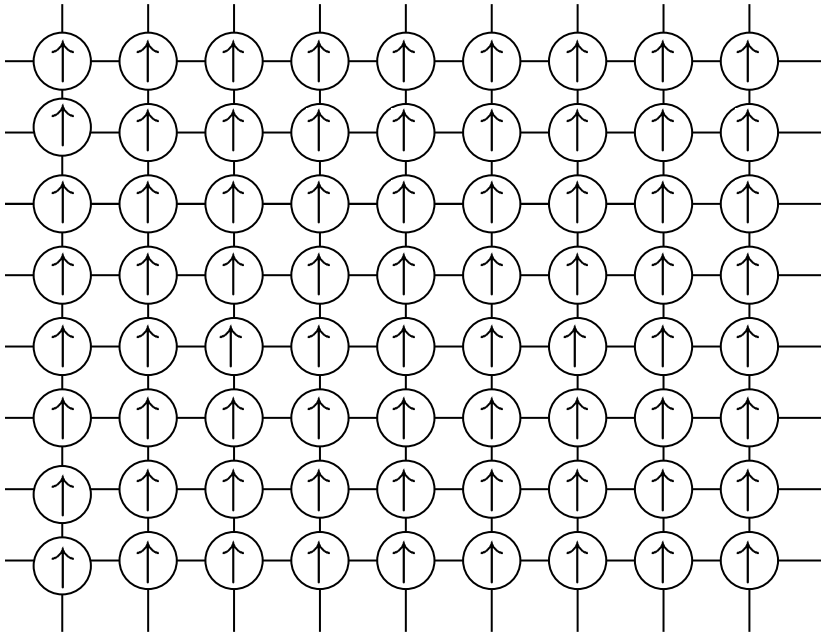
$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



A Dynamic Ising Model



Ising model

spins on a lattice $s_i \in \{+1, -1\}$

$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i$$

Energetically favorable to align

Order parameter:

$$M = \sum_i s_i$$

Dynamic Ising model (Metropolis update)

interaction with environment causes spins to flip

for every spin

if flipping spin would decrease energy, flip spin

if flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

Metropolis Monte Carlo

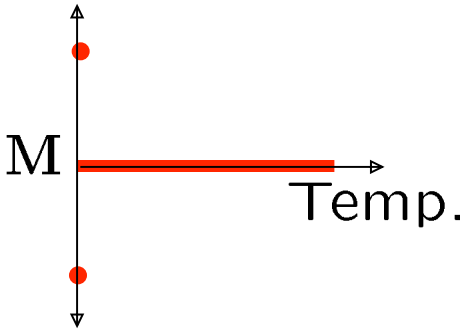
Storage and Manipulation

Compare 1D and 2D Ising models at different temperatures for
a) storage of information
b) manipulation of information

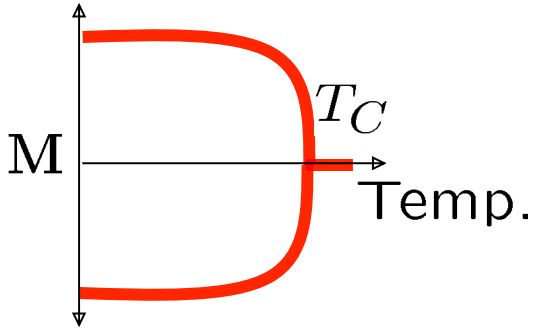
1st attempt:

Storage in thermal state (infinite time - ensemble is thermal)

1D Ising



2D Ising



Criticism: (1) thermodynamic limit taken (N infinite)
(2) what if relaxation to equilibrium takes a long time?

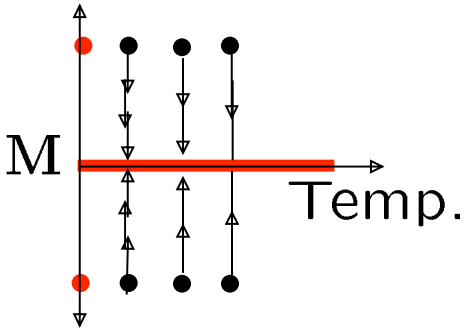
Storage



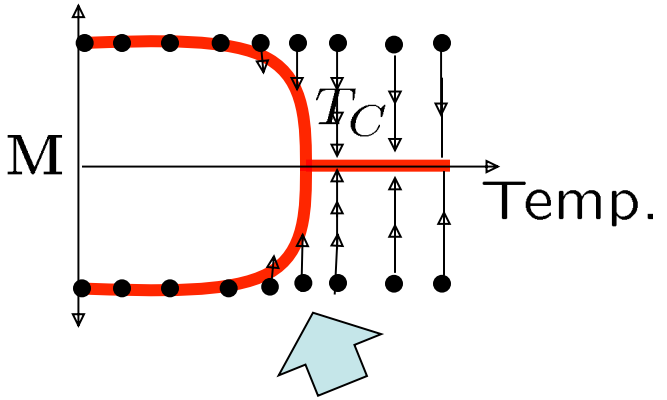
2nd attempt:

Relaxation to thermal

1D Ising



2D Ising

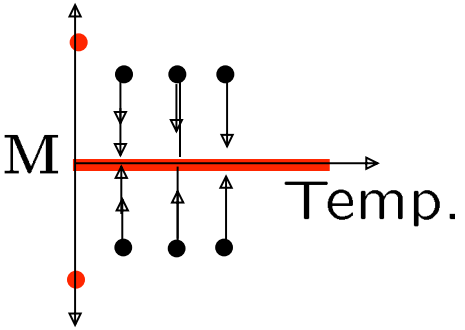


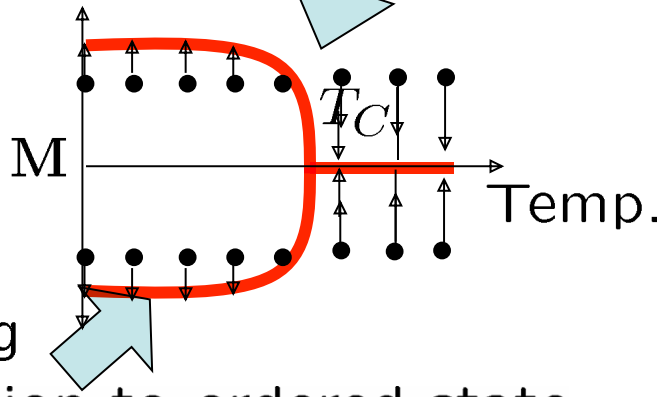
critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

decay rate $\propto e^{-\beta\Delta}$

Imperfect preparation





self correcting

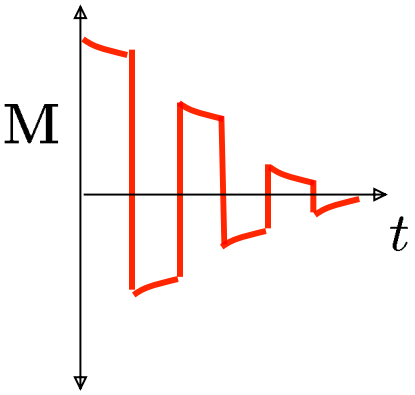
fast relaxation to ordered state

Manipulation

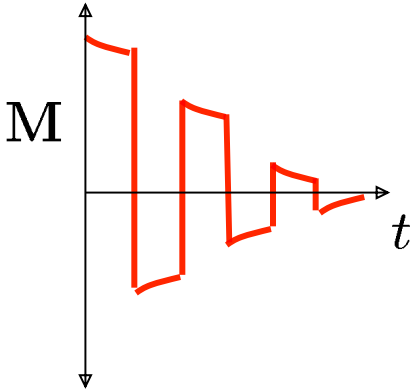


Flipping spins is imperfect

1D Ising

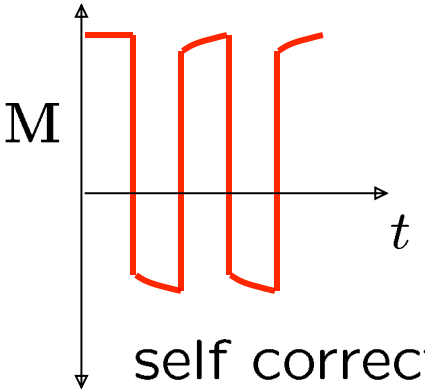


2D Ising



$T > T_C$

1D Ising
- temperature less than gap
- manipulation error must be small

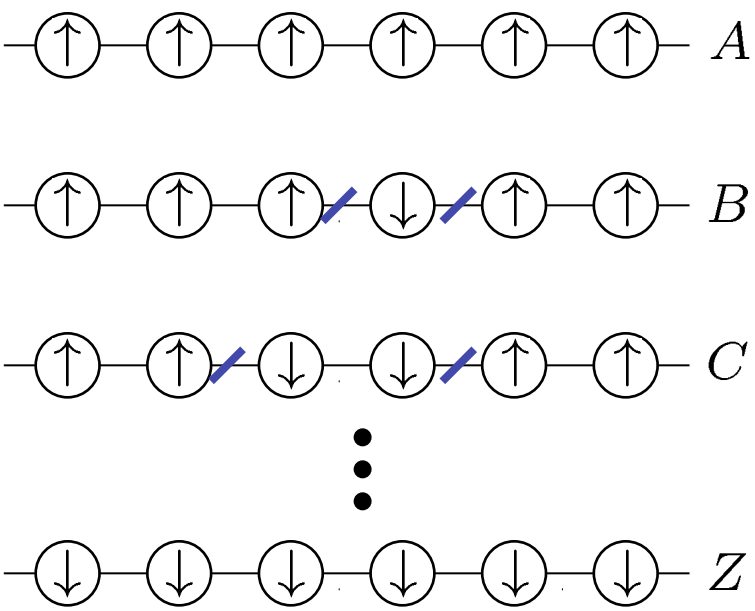


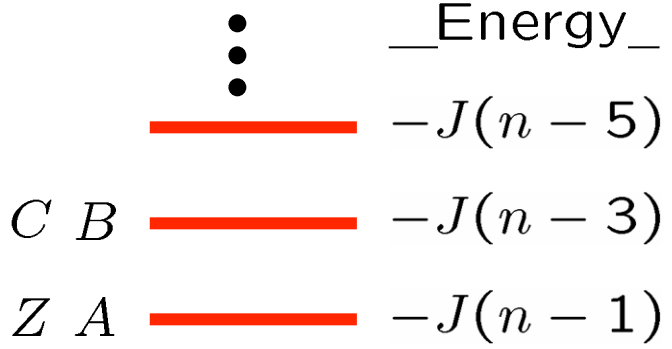
$T < T_C$

2D Ising
- temperature less than critical temp
- self correcting

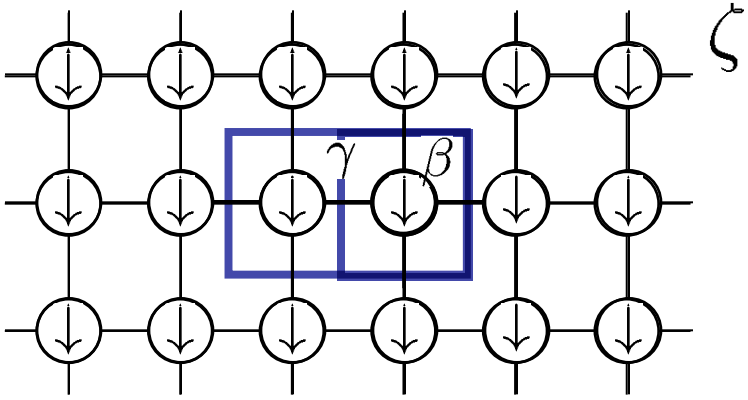
self correcting

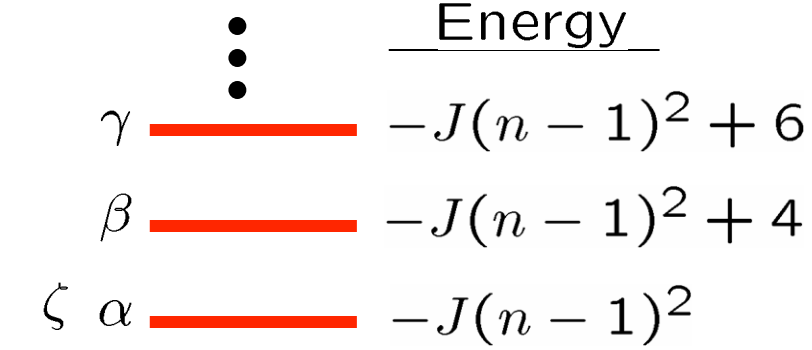
Self Fixing





Excitations free pointlike.
Energy cost constant.
disordering entropy $\gg$ energy



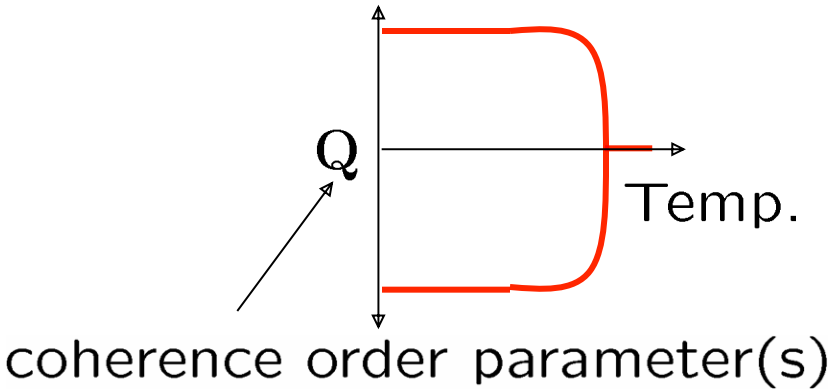


Excitations stringlike.
Energy cost grows like perimeter.
energy $\gg$ disordering entropy

The Quantum Hard Drive?

(Kitaev)

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.

$Q_1 \rightarrow -Q_1$

macro local operations

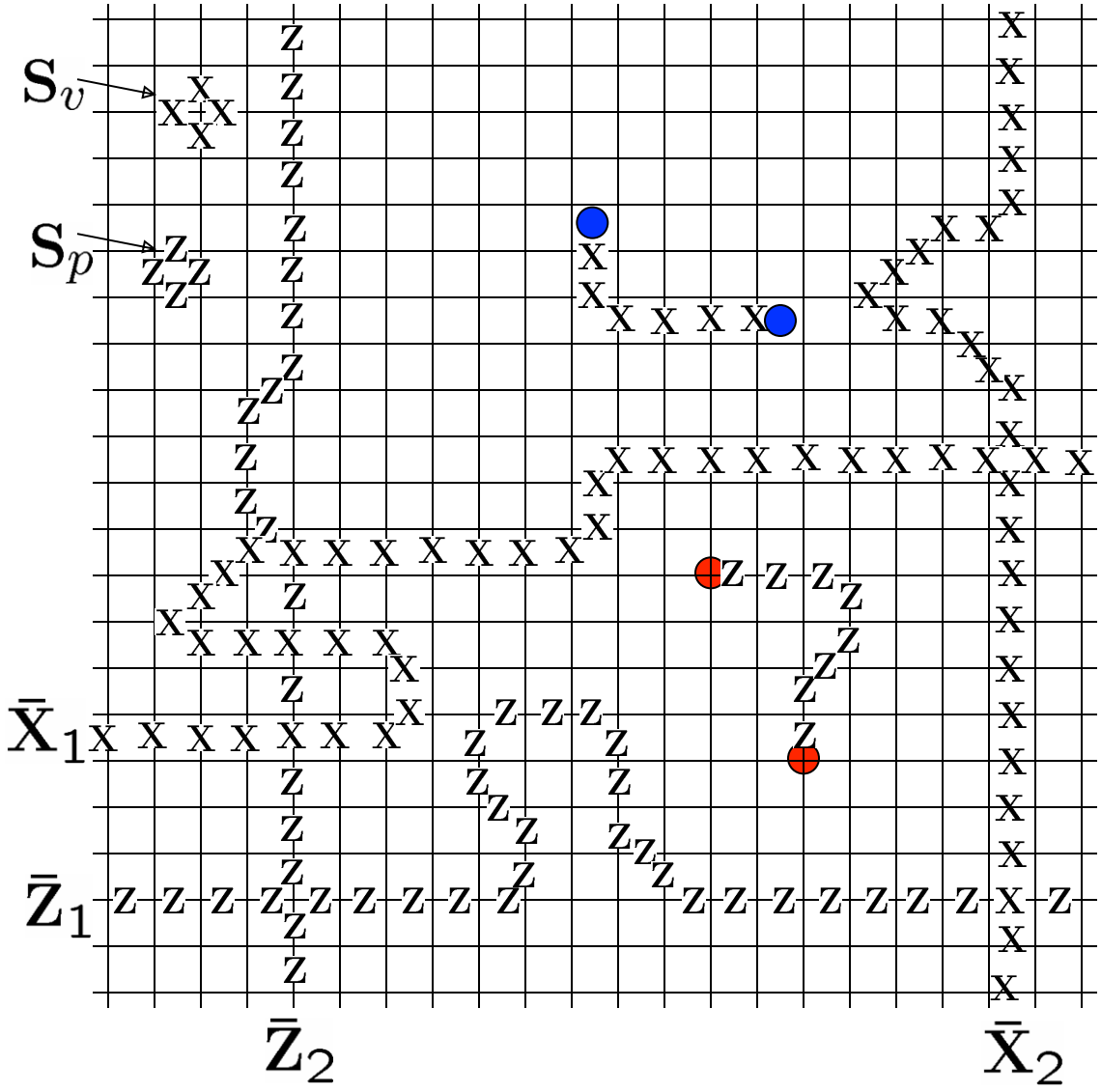
2. Accessible Fault-Tolerant Operations

"self-correcting"

$\langle G_1, \ldots, G_{n^k} \rangle \cong U(c^n)$

3. Universality

Wherefore Art Toric Codes?



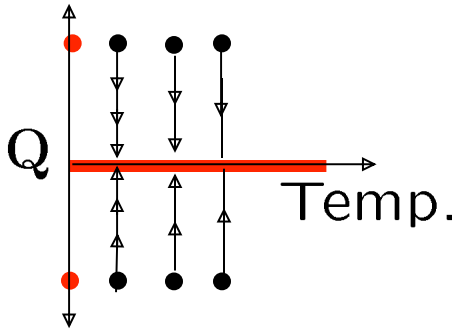
$$\mathbf{S}_p = \sum_{e \in p} \mathbf{Z}_e$$
$$\mathbf{S}_v = \sum_{e \in v} \mathbf{X}_e$$

$$\mathbf{H} = -\frac{\Delta}{2}\left(\sum_p \mathbf{S}_p + \sum_v \mathbf{S}_v\right)$$

energy gap $= \Delta$

can correct almost any $O(L^2)$ qubit errors (percolation)

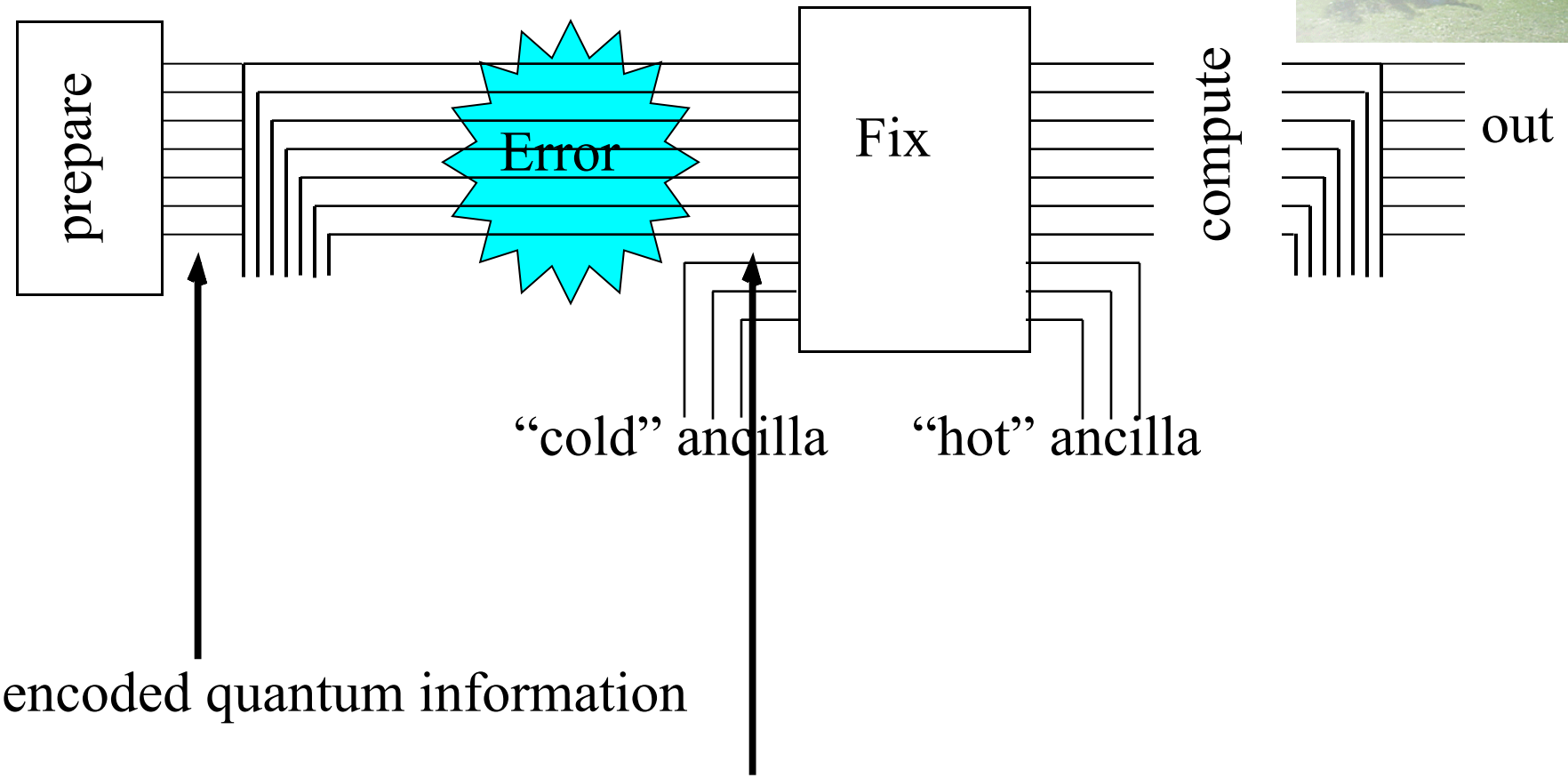
natural evolution is not self correcting (we have to work in 2D)



4D Toric code should be self correcting!

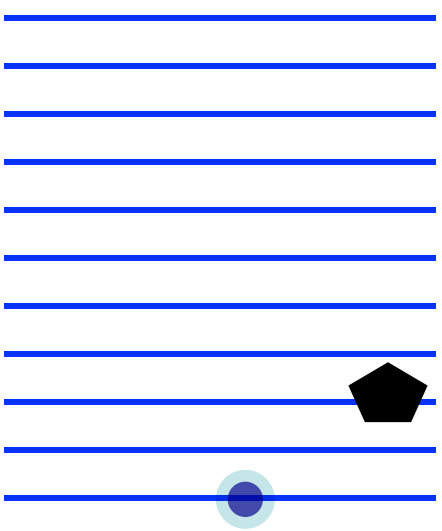
Old School





How do we reveal erred quantum information?

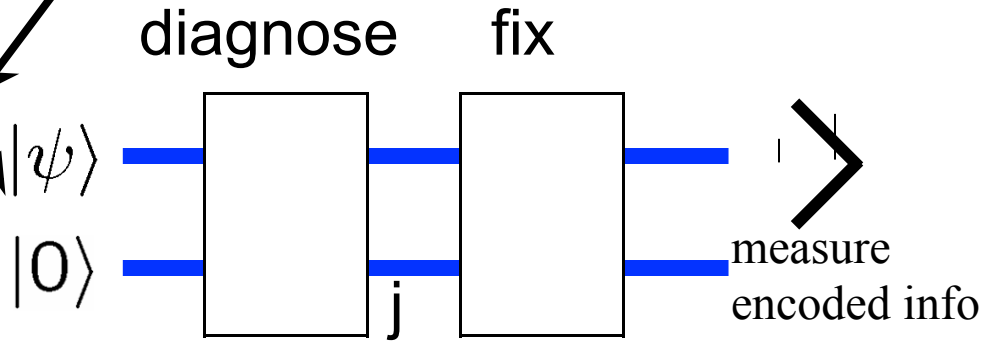
Quantum Error Correcting Order Parameter



$|\psi\rangle$

ρ_{error}

The quantum information is still here. How do we see it?



Encoded quantum information

$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\mathbf{Q}_s = \sum_j \mathbf{L}_s \mathbf{F}_j \mathbf{D}_j$$

$\mathbf{L}_s$ logical encoded operation

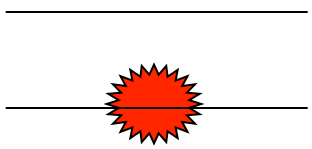
$\mathbf{D}_j$ projector onto jth errored subspace

$\mathbf{F}_j$ fixing operator for jth errored subspace

Subsystem Codes

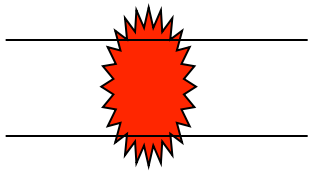
The most generic error correction encoding of quantum information is not into a subspace, but into a subsystem

trivial example:



two qubits, one is errored
info in first is not destroyed

slightly less trivial example:



$$|z_1, z_2\rangle \rightarrow |z_1 \oplus z_2, z_2\rangle$$

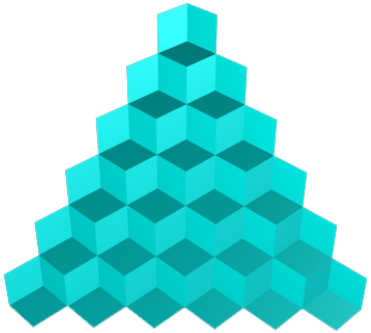
parity qubit is never erred

$$\mathbf{A}_1 = \sqrt{1-p}\mathbf{I} \otimes \mathbf{I},\ \mathbf{A}_2 = \sqrt{p}\mathbf{Z} \otimes \mathbf{Z}$$

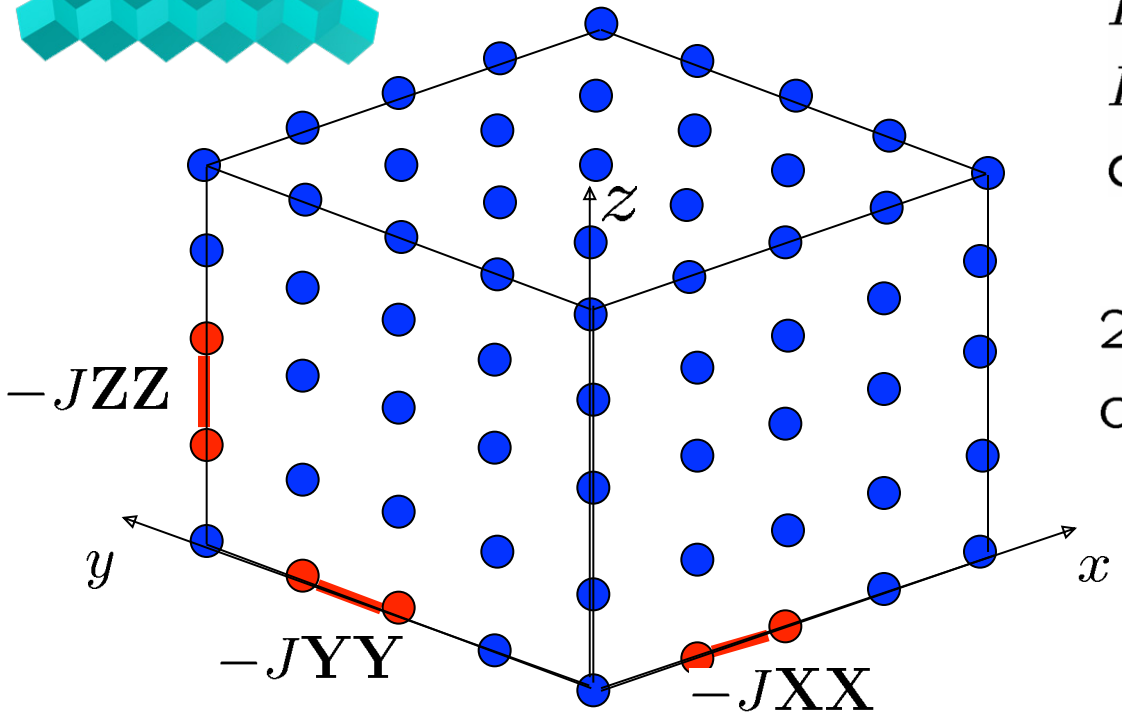
Generically, we can encode information into the a tensor factor of some subspace:

$$\mathcal{H} = \oplus_\mu \otimes_\nu \mathcal{H}_{\mu\nu}$$

Error correction condition is NOT that a subspace is recoverable, bu
that information in a subsystem is recoverable (KLV)



Into the Third Dimension



$L \times L \times L$ cubic lattice

L odd

qubits on vertices

2 qubit interactions along cartesian directions

Some of you may remember my old favorite Hamiltonian:

$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

Fudgy arguments about why persistent coherence…

How to proceed: simulate! But this systems has a “sign” problem.

Quantum Monte Carlo

Thermal ensemble:

$$\boldsymbol{\rho}_{th} = \frac{e^{-\beta \mathbf{H}}}{Z}, \quad Z = \mathrm{Tr}\left[e^{-\beta \mathbf{H}}\right]$$

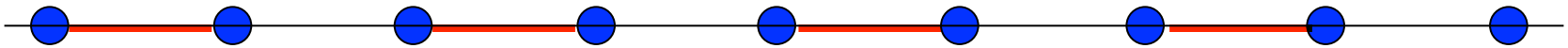
We want to understand coherence order parameters

$$\langle \mathbf{Q}_s \rangle = \mathrm{Tr}\left[\boldsymbol{\rho}_{th} \mathbf{Q}_s\right]$$

Since diagonalizing $\mathbf{H}$ is hard, how do we perform monte carlo?

Example: anisotropic Heisenberg chain

$$\mathbf{H}_{ex} = \sum_{i=1}^{n} \left(J_x \mathbf{X}_i \mathbf{X}_{i+1} + J_y \mathbf{Y}_i \mathbf{Y}_{i+1} + J_z \mathbf{Z}_i \mathbf{Z}_{i+1} \right)$$



$$\mathbf{H}_{ex} = \mathbf{H}_o + \mathbf{H}_e$$

$$\mathbf{H}_o = \sum_{i=1|i \text{ odd}}^{n} \left(J_x \mathbf{X}_i \mathbf{X}_{i+1} + J_y \mathbf{Y}_i \mathbf{Y}_{i+1} + J_z \mathbf{Z}_i \mathbf{Z}_{i+1} \right)$$

$$\mathbf{H}_e = \sum_{i=1|i \text{ even}}^{n} \left(J_x \mathbf{X}_i \mathbf{X}_{i+1} + J_y \mathbf{Y}_i \mathbf{Y}_{i+1} + J_z \mathbf{Z}_i \mathbf{Z}_{i+1} \right)$$

Note fixed i terms commute for $\mathbf{H}_o$ and $\mathbf{H}_e$

QMC Example Continued

$$Z = \mathrm{Tr}\left[e^{-\beta \mathbf{H}}\right] = \sum_n \langle n| \left(e^{-\frac{\beta}{N}(\mathbf{H}_o + \mathbf{H}_e)}\right)^N |n\rangle$$

$$= \lim_{N\to\infty} \prod_{k=1,k \text{ odd}}^{2N} \sum_{|n_k\rangle} \langle n_k|e^{-\frac{\beta}{N}\mathbf{H}_o}|n_{k+1}\rangle\langle n_{k+1}|e^{-\frac{\beta}{N}\mathbf{H}_e}|n_{k+2}\rangle$$

Fix some (convenient) basis for the $|n_k\rangle$. For example, the $\boldsymbol{\sigma}_z$ basis for all qubits.

$$= \lim_{N\to\infty} \sum_{\{|s_k\rangle\}} \prod_{k=1,k \text{ odd}}^{2N} \langle s_k|e^{-\frac{\beta}{N}\mathbf{H}_o}|s_{k+1}\rangle\langle s_{k+1}|e^{-\frac{\beta}{N}\mathbf{H}_e}|s_{k+1}\rangle$$

Sum over classical configurations $\{|s_k\rangle\}$ $\qquad \epsilon = \frac{\beta}{N}$

$$\langle s_k|e^{-\epsilon \mathbf{H}_o}|s_{k+1}\rangle = \langle s_k| \prod_{i=1,i \text{ odd}} e^{-\epsilon \mathbf{H}_i}|s_{k+1}\rangle$$

$$\mathbf{H}_i = \left(J_x \mathbf{X}_i \mathbf{X}_{i+1} + J_y \mathbf{Y}_i \mathbf{Y}_{i+1} + J_z \mathbf{Z}_i \mathbf{Z}_{i+1}\right)$$

$$\langle s_k|e^{-\epsilon \mathbf{H}_o}|s_{k+1}\rangle = \prod_{i=1,i \text{ odd}} \langle (s_k)_i (s_k)_{i+1}|e^{-\epsilon \mathbf{H}_i}|(s_{k+1})_i (s_{k+1})_{i+1}\rangle$$

QMC III

QUICK MED CLAIMS

$$\langle (s_k)_i (s_k)_{i+1} | e^{-\epsilon \mathbf{H}_i} | (s_{k+1})_i (s_{k+1})_{i+1} \rangle =$$

$$| (s_{k+1})_i (s_{k+1})_{i+1} \rangle$$

$$| (s_k)_i (s_k)_{i+1} \rangle \begin{bmatrix} e^{-\epsilon J_z} \cosh(\epsilon J_+) & 0 & 0 & e^{-\epsilon J_z} \sinh(\epsilon J_+) \\ 0 & e^{\epsilon J_z} \cosh(\epsilon J_-) & e^{\epsilon J_z} \sinh(\epsilon J_-) & 0 \\ 0 & e^{\epsilon J_z} \sinh(\epsilon J_-) & e^{\epsilon J_z} \cosh(\epsilon J_-) & 0 \\ e^{-\epsilon J_z} \sinh(\epsilon J_+) & 0 & 0 & e^{-\epsilon J_z} \cosh(\epsilon J_+) \end{bmatrix}$$

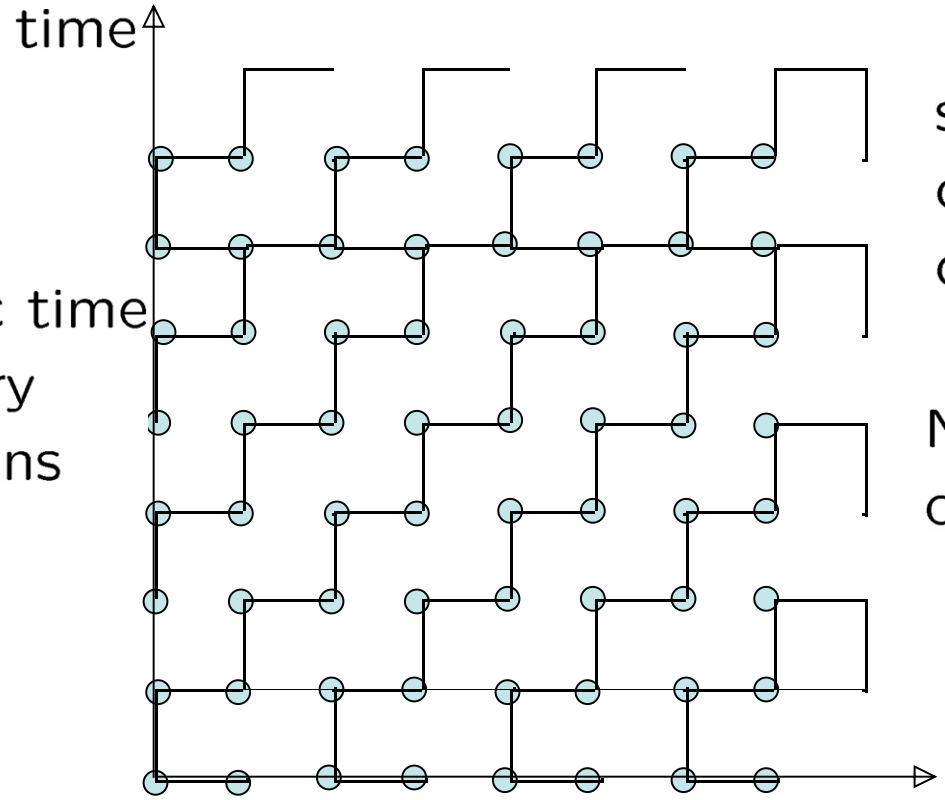
$$J_\pm = J_y \pm J_x$$

0	0	0	0	0	1	0	1
0	0	1	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	0	0	0	1	1	0

$e^{-\epsilon J_z} \cosh(\epsilon J_+)$ $e^{-\epsilon J_z} \sinh(\epsilon J_+)$ $e^{+\epsilon J_z} \sinh(\epsilon J_-)$ $e^{+\epsilon J_z} \cosh(\epsilon J_-)$

QMC III

$$Z = \lim_{N\to\infty} \sum_{\{|s_k\rangle\}} \prod_{k=1,k\ \text{odd}}^{2N} \langle s_k|e^{-\frac{\beta}{N}\mathbf{H}_o}|s_{k+1}\rangle\langle s_{k+1}|e^{-\frac{\beta}{N}\mathbf{H}_e}|s_{k+1}\rangle$$



Periodic time boudnary conditions

sum over classical configurations of $d+1$ dimensional lattice

Nonzero contribution only on colored plaquettes

Partition function on d dim quantum system

Partition-like function on $d+1$ dim classical system

Partition-Like = Sign Problem

$$\begin{bmatrix} e^{-\epsilon J_z}\cosh(\epsilon J_+) & 0 & 0 & e^{-\epsilon J_z}\sinh(\epsilon J_+) \\ 0 & e^{\epsilon J_z}\cosh(\epsilon J_-) & e^{\epsilon J_z}\sinh(\epsilon J_-) & 0 \\ 0 & & & 0 \\ e^{-\epsilon J_z}\sinh(\epsilon J_+) & & & e^{-\epsilon J_z}\cosh(\epsilon J_+) \end{bmatrix}$$

If J_- or J_+ is negat

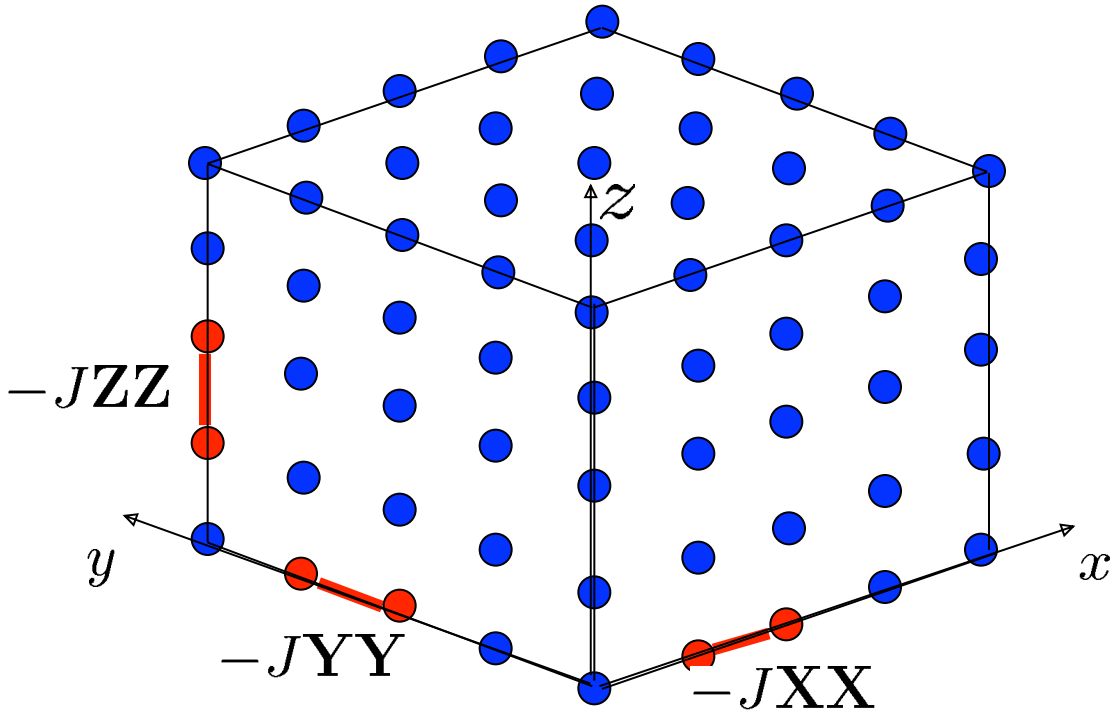
Partition functior ability interp

If there are negative bilities!

Sometimes sign pro hange.



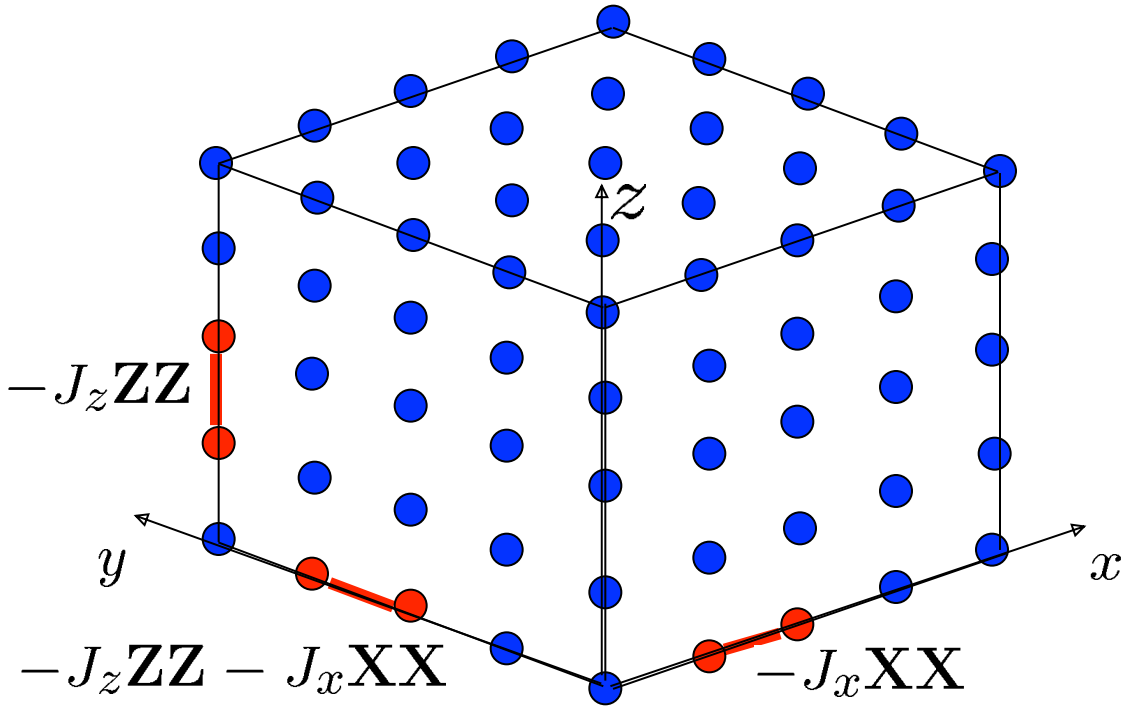
Back to the Third Dimension



$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

If we use $Z_{i,j,k}$ basis, $\mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k}$ gives sign problem

The Third Dimension: Take II

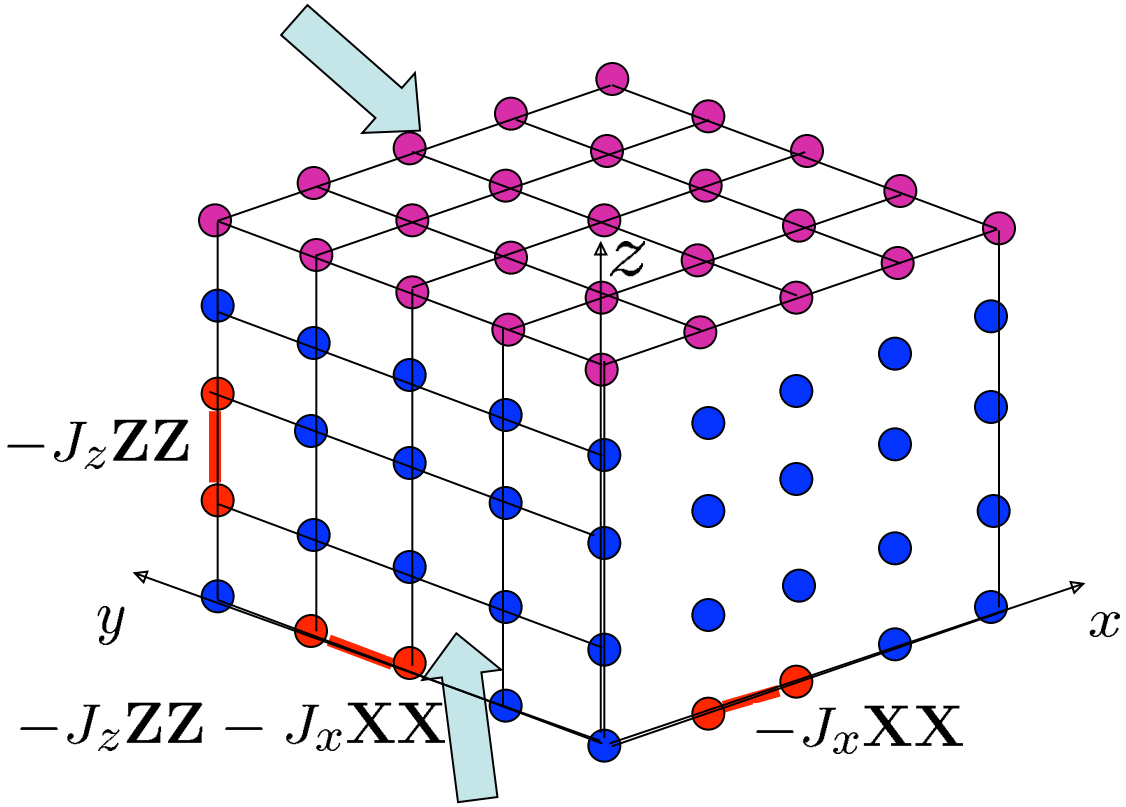


$$\mathbf{H} = -\sum_{ijk} J_x(\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{X}_{i,j,k}\mathbf{X}_{i,j+1,k}) - \sum_{ijk} J_z(\mathbf{Z}_{i,j,k} + \mathbf{Z}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}))$$

Stacks of Ising

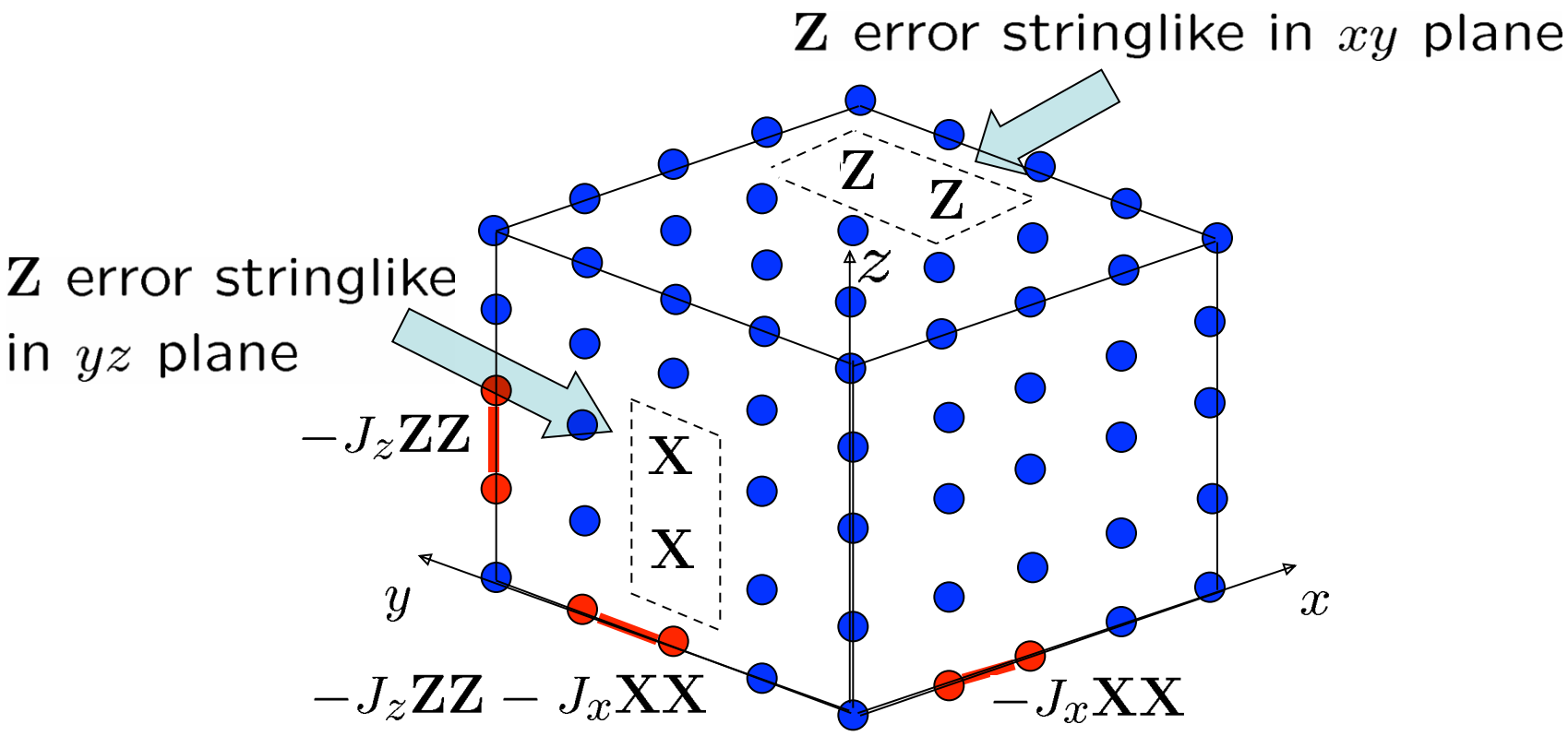


xy planes are 2D XX ising



yz planes are 2D ZZ ising

Me Like-y



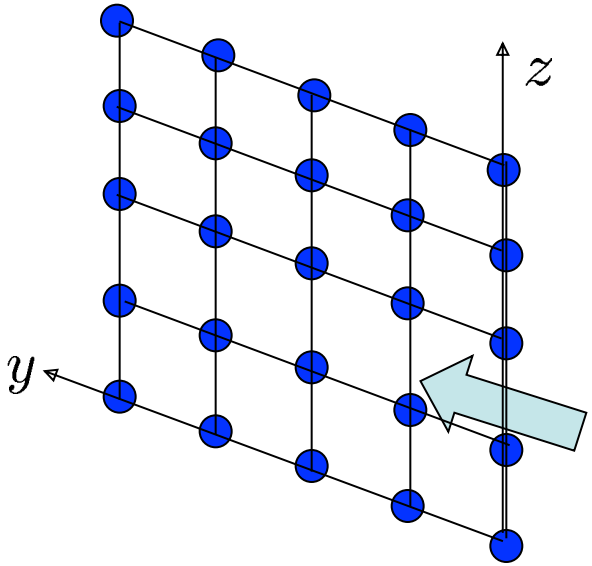
$\{\mathbf{XX}, \mathbf{ZI}\} = 0,$ etc.

Group of Bonds (Bobby and Barry)

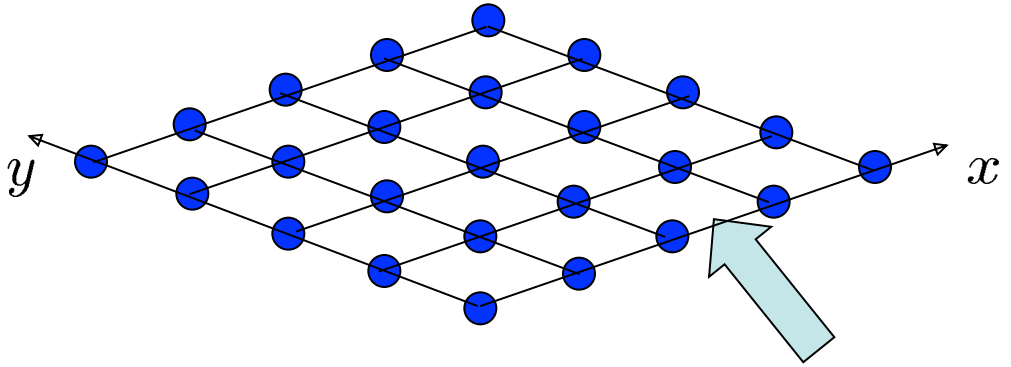
$$\mathbf{H} = -\sum_{ijk} J_x(\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{X}_{i,j,k}\mathbf{X}_{i,j+1,k}) - \sum_{ijk} J_z(\mathbf{Z}_{i,j,k} + \mathbf{Z}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}))$$

Examine the group generated by the bonds

$$\mathcal{A} = \langle \mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k}, \mathbf{X}_{i,j,k}\mathbf{X}_{i,j+1,k}, \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j+1,k}, \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1} \rangle$$

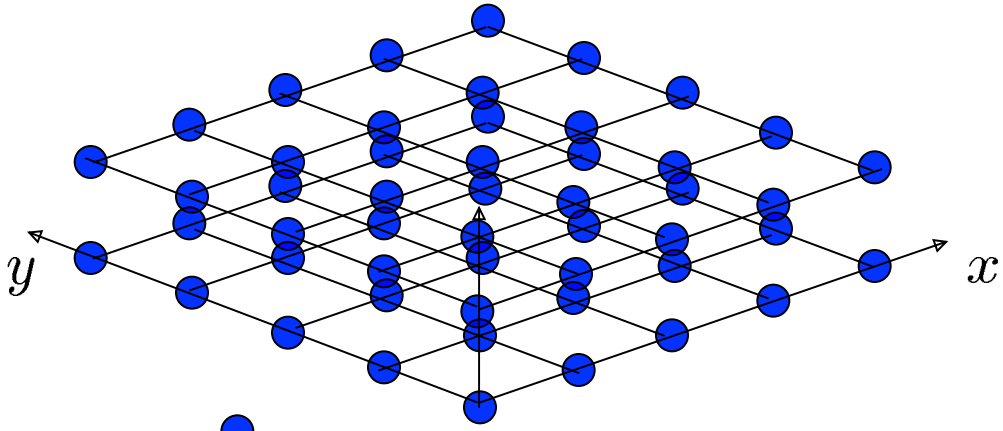


Even number of Zs in yz planes



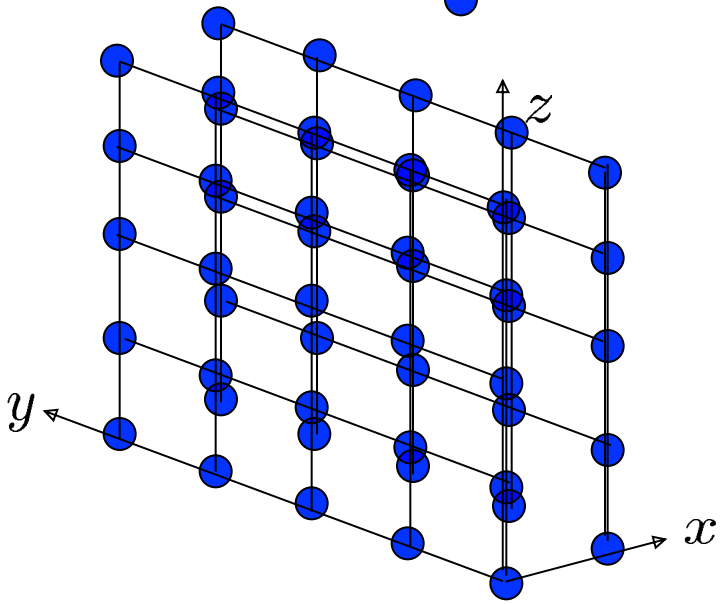
Even number of Xs in xy planes

Abelian Invariant Subgroup



Double planes of $\mathbf{Z}$s

$$\mathbf{E}_k = \prod_{i,j} \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}$$



Double yz planes of $\mathbf{X}$s

$$\mathbf{F}_i = \prod_{j,k} \mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k}$$

$\mathcal{S} = \langle \mathbf{E}_k, \mathbf{F}_i \rangle \subset \mathcal{A}$ $\quad$ $\mathcal{S}$ is abelian

$$\mathbf{SA} = \mathbf{AS},\ \mathbf{A} \in \mathcal{A},\ \mathbf{S} \in \mathcal{S}$$

$\mathcal{S}$ is an abelian invariant subgroup of $\mathcal{A}$

Painful Constructio



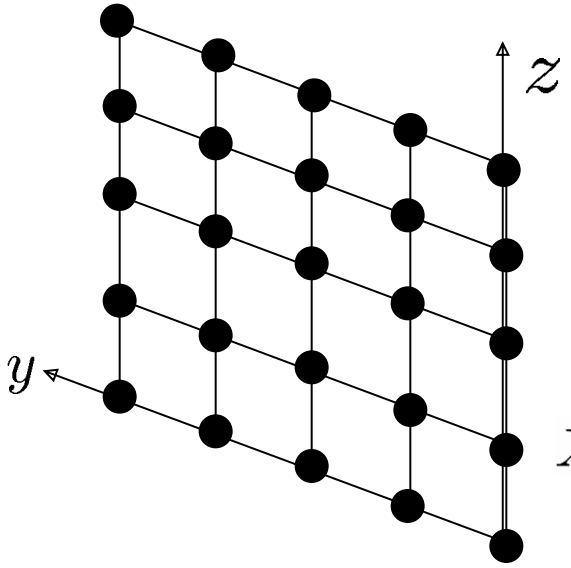
Group elements can be written $\mathcal{SP} = \mathcal{PS}$

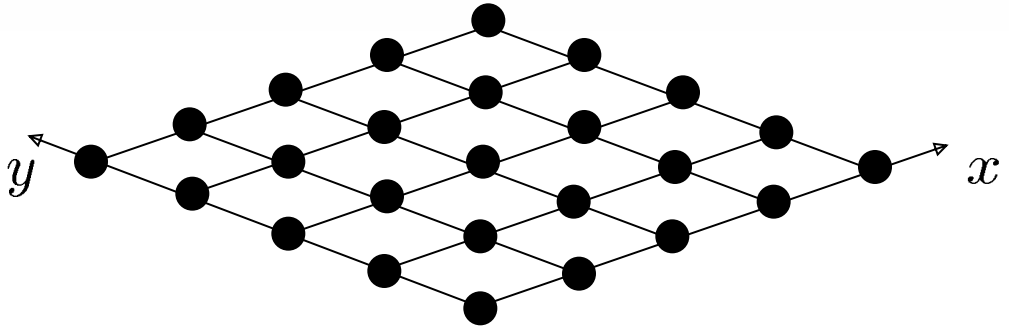
$\mathcal{S}$ is the abelian invariant subalgebra

$\mathcal{P}$ is the Pauli group on $n^3 - 2n + 1$ qubits

The construction for the Pauli group is beyond ugly

But $n^3 - (n^3 - 2n + 1 + 2(n-1)) = 1$, so there must be one encoded qubit

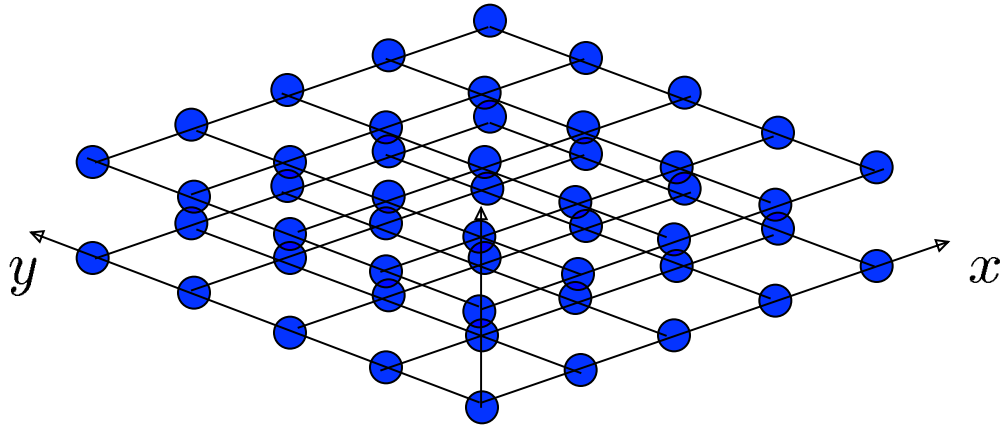




$\bar{Z}$ =Product of n^2 **Z**s in xy plane

$\bar{X}$ =Product of n^2 **X**s in yz plane

The Subsystem Code



Double planes of $\mathbf{Z}$s

$\mathbf{E}_k = \prod_{i,j} \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}$

Determines if odd or even number of $\mathbf{X}$ errors in double xy plane

Each xy plane has even or odd $\mathbf{X}$ errors

$e,\ e,\ e,\ e,\ e,\ \ldots,\ e \quad \mathbf{E}_k = e$

$error$

$e,\ e,\ o,\ e,\ e,\ \ldots,\ e$

$\mathbf{E}_2 = o \quad \mathbf{E}_3 = o$

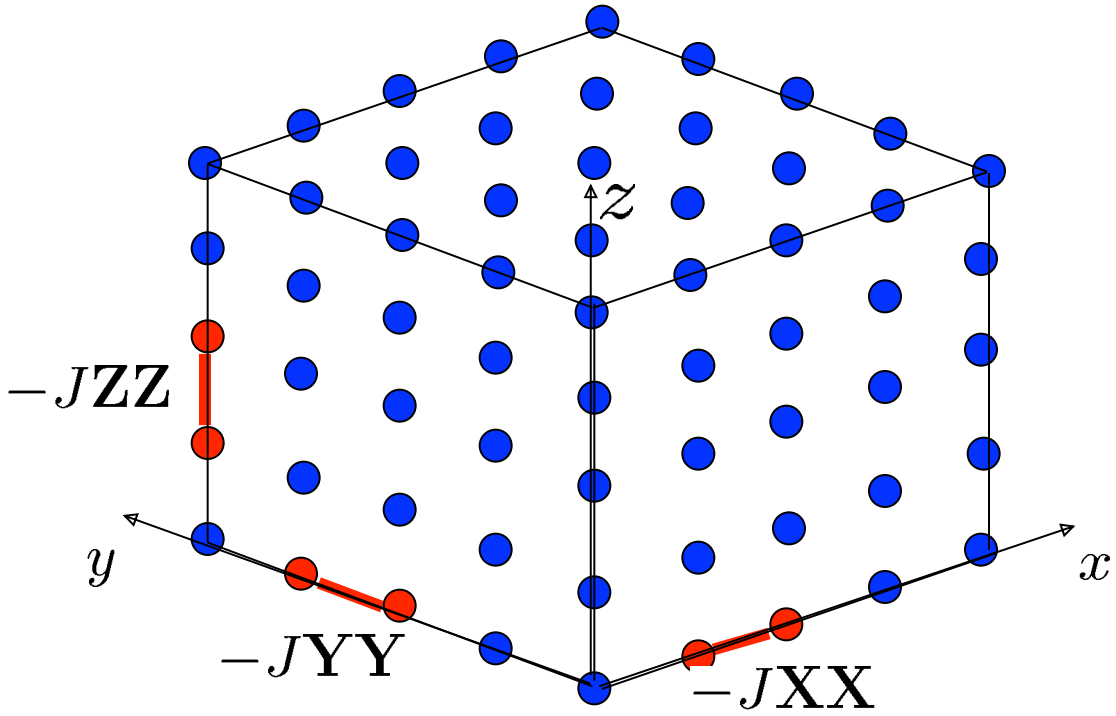
Using $\mathbf{E}_k$ measurements we can fix $(n-1)/2$ odds to evens

If all evens, the $\bar{\mathbf{Z}}$ has been left invariant!

But evens are elements of $\mathcal{P}$!

Same for yz planes

Status



doubly degenerate energy levels

subsystem code

can write down order parameter for subsystem code

no sign problem

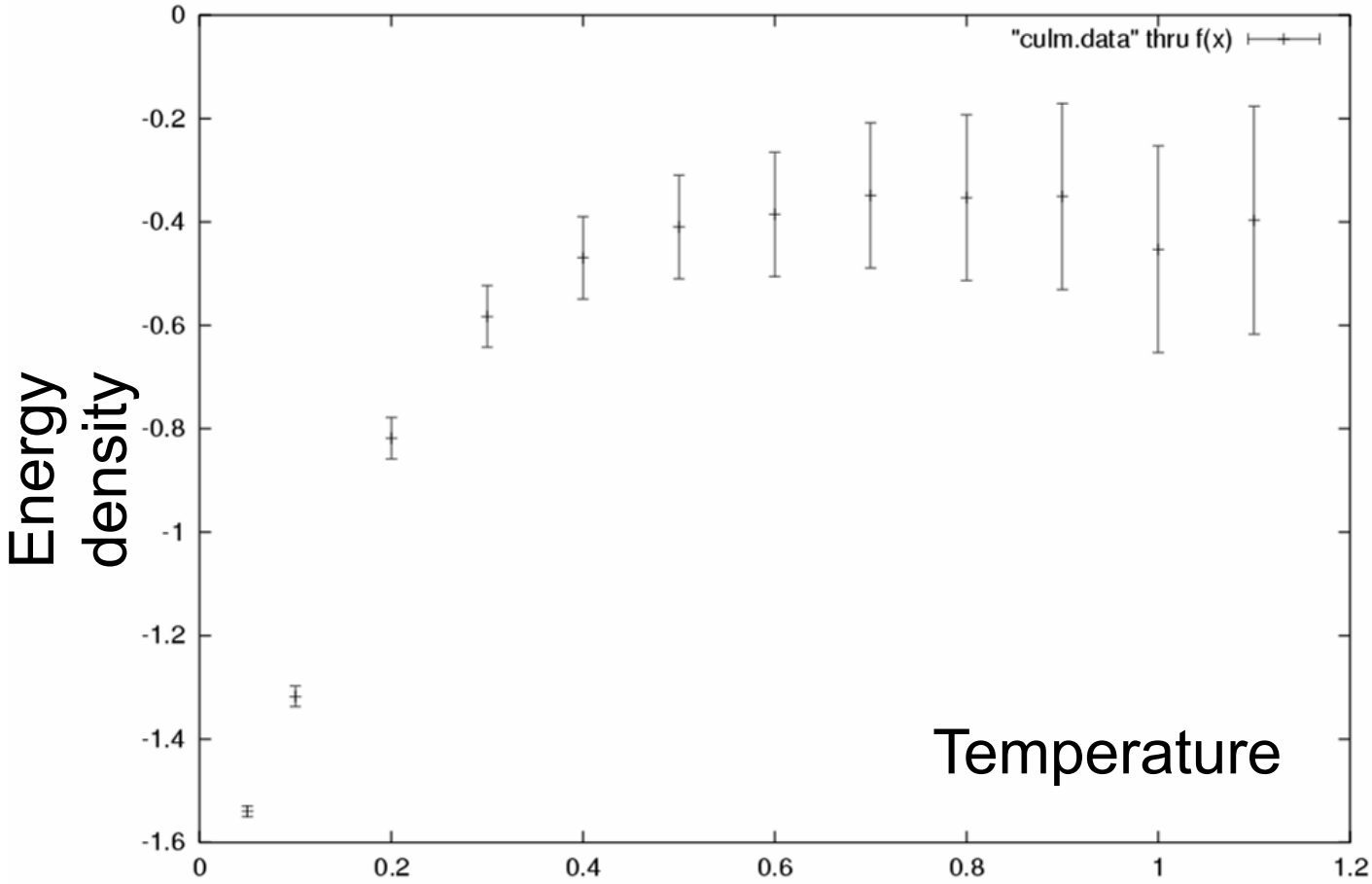
intuition of stringlike excitations

Loop Algorithm

Quantum Monte Carlo method for systems with no sign problem

Cluster-like algorithm with improved estimators

Allows measurement of off-diagonal operators

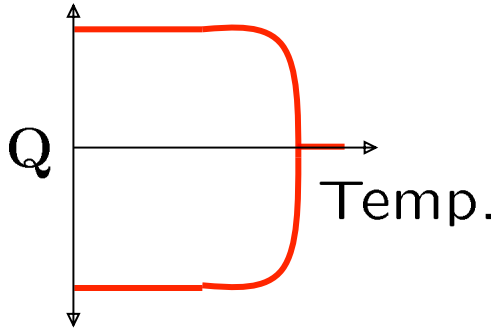


$J_x = J_z$

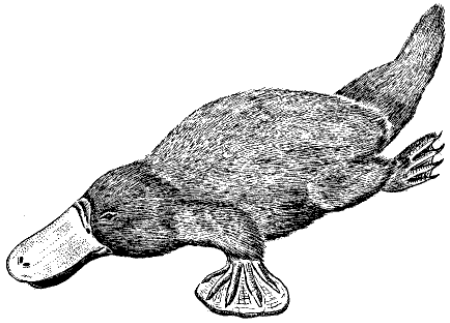
$L = 7$, $T = 21$, 100000 MC steps

Units are $J_x = J_z = 1$

The question!



Fin



Platyputor